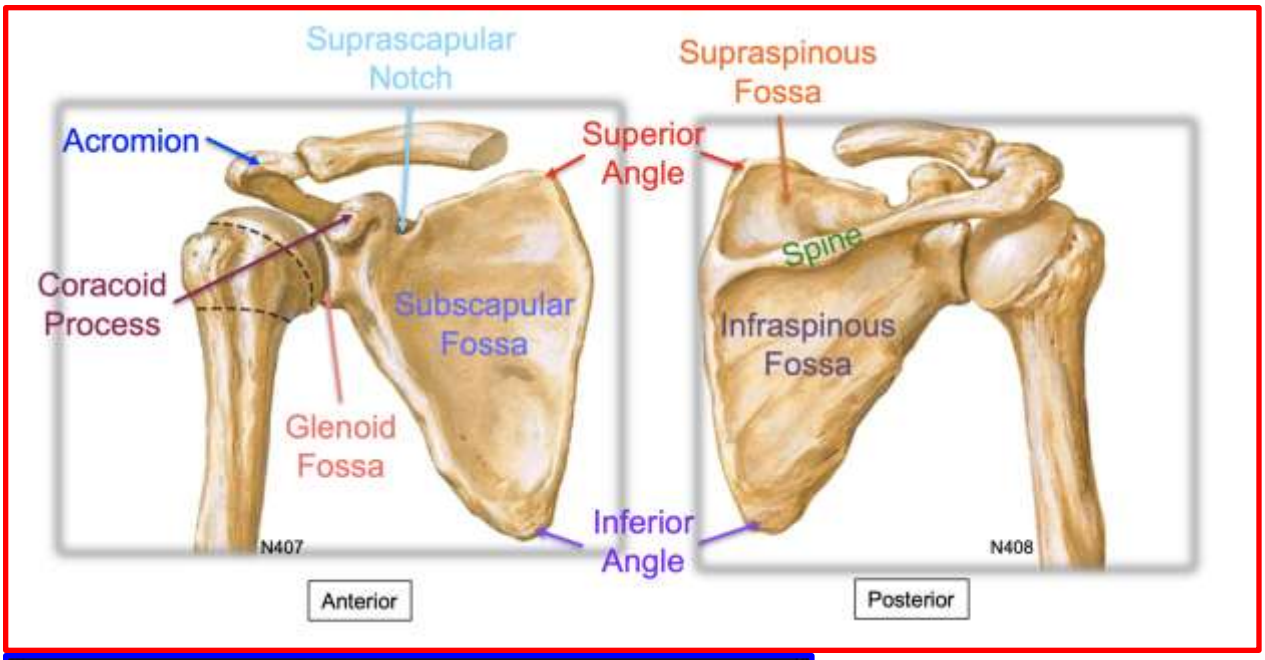
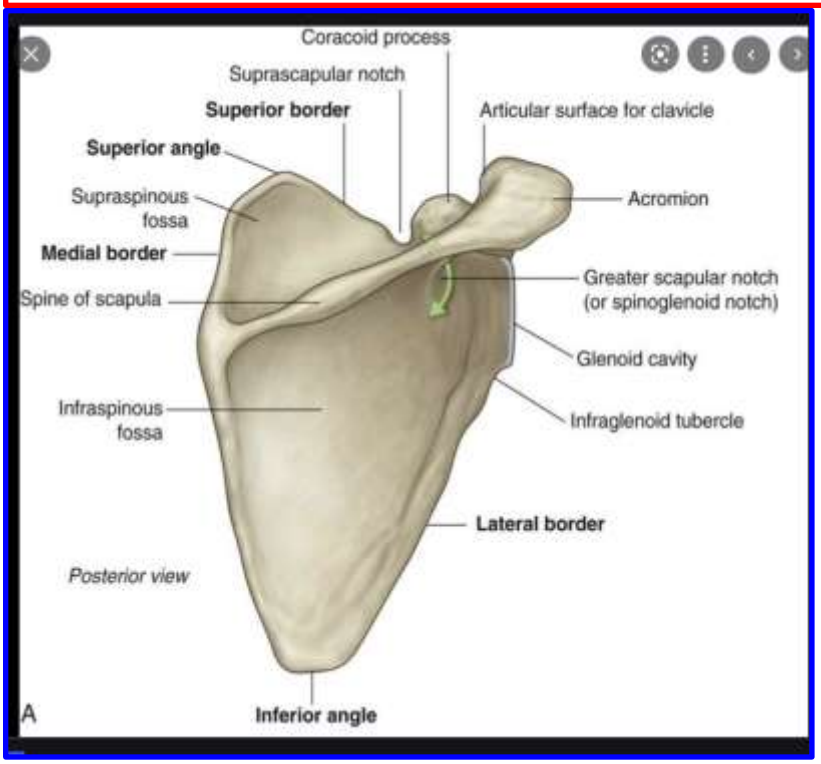
MSK: Clinical Anatomy of the Proximal Upper Extremity

You will have two labs to complete the entirety of this lab. Work diligently to get through the majority of A-E during the first lab. Whatever remains can be completed with F-I during the second lab.

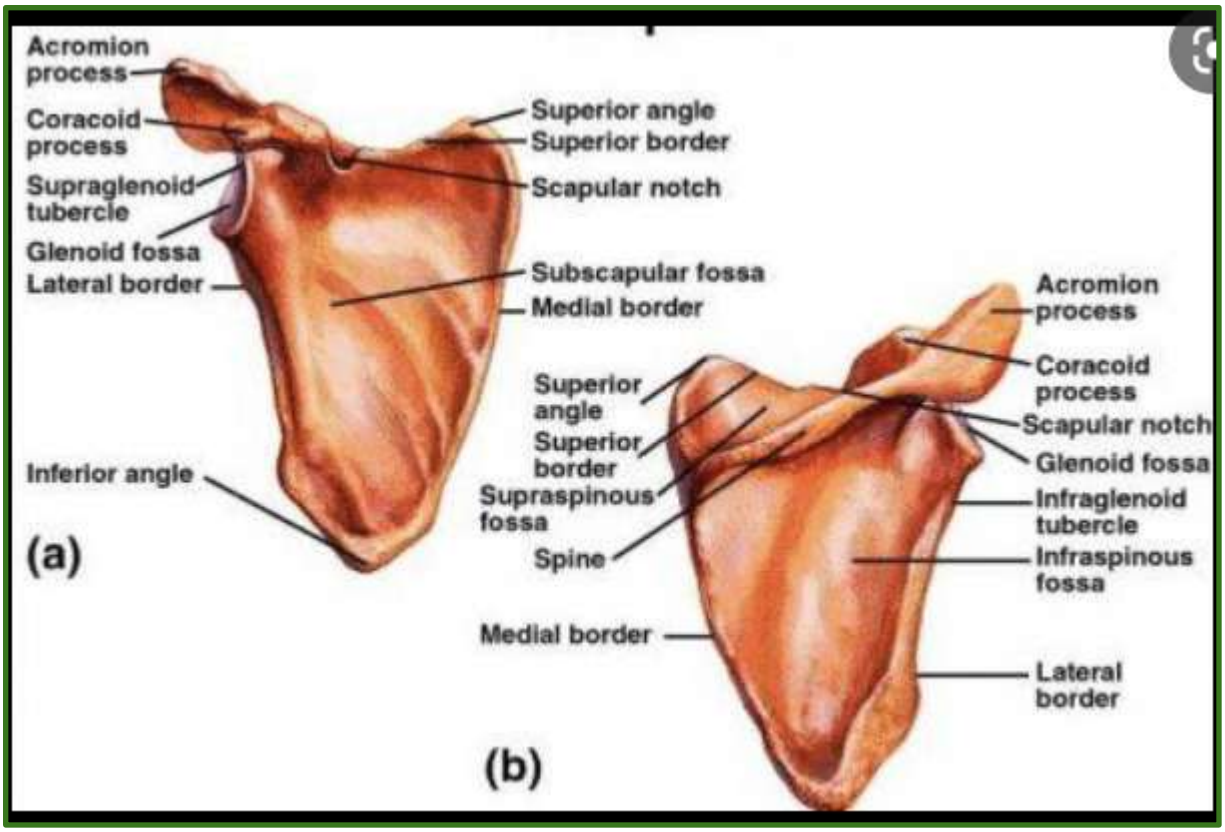
A. Scapular Region of Shoulder:

1. Locate The Following On The Bony Scapula:
    a. **Spine**
    b. **Coracoid Process**
    c. **Glenoid Fossa**
    d. **Subscapular, Supraspinous,** and **Infraspinous Fossae**
    e. **Suprascapular Notch**
    f. **Spinoglenoid (Great Scapular Notch) Notch**
    g. **Supraglenoid** and **Infraglenoid Tubercles**
    h. **Lateral, Superior** and **Medial (Vertebral) Borders**
    i. **Lateral, Superior** and **Inferior Angles**



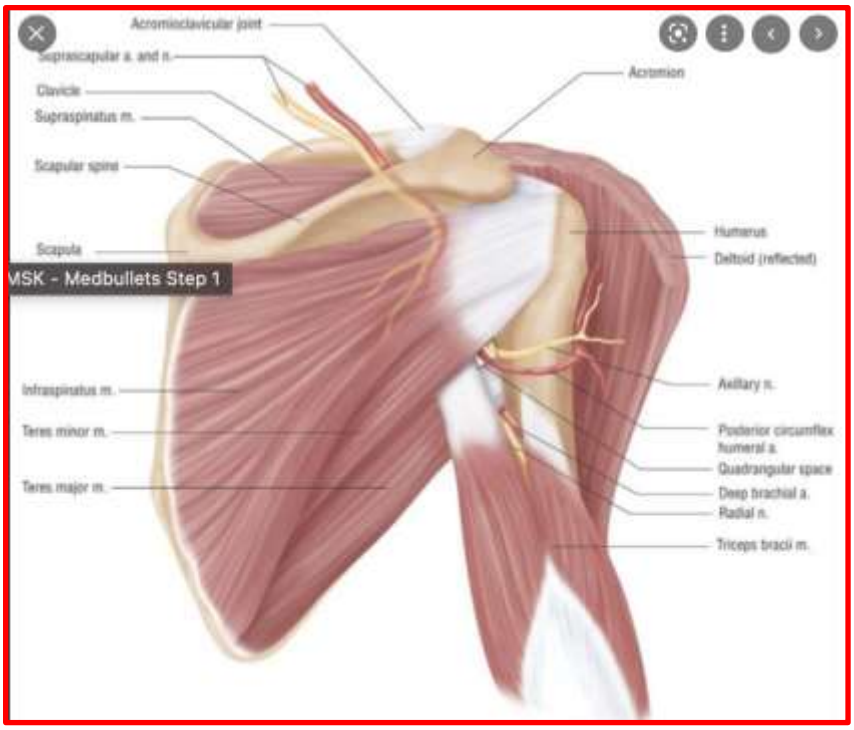


^**Spinoglenoid notch**

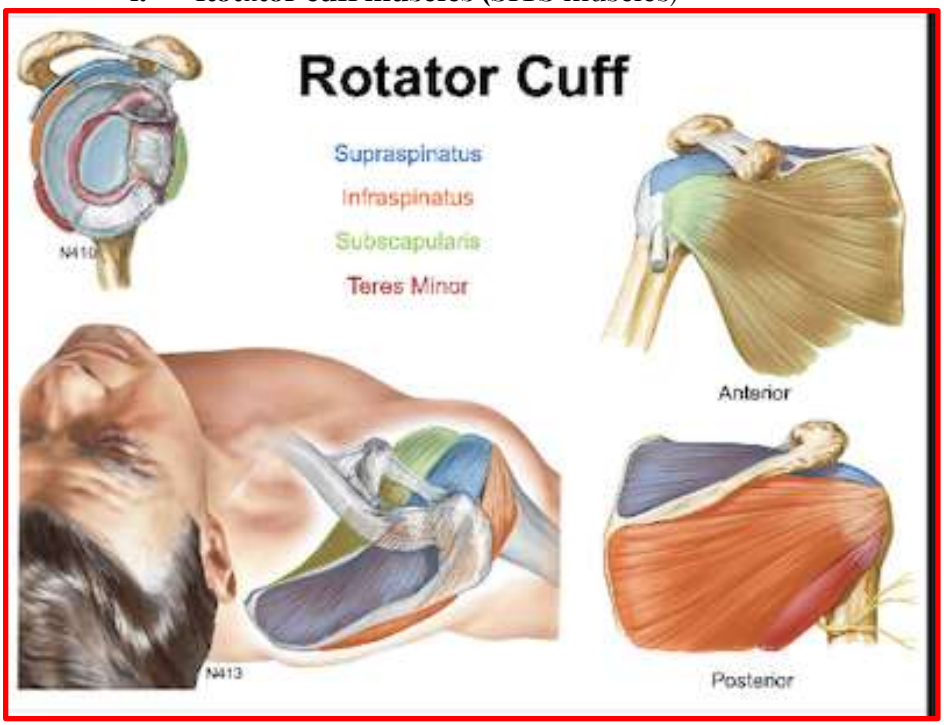


- **Supraglenoid, infraglenoid tubercle**

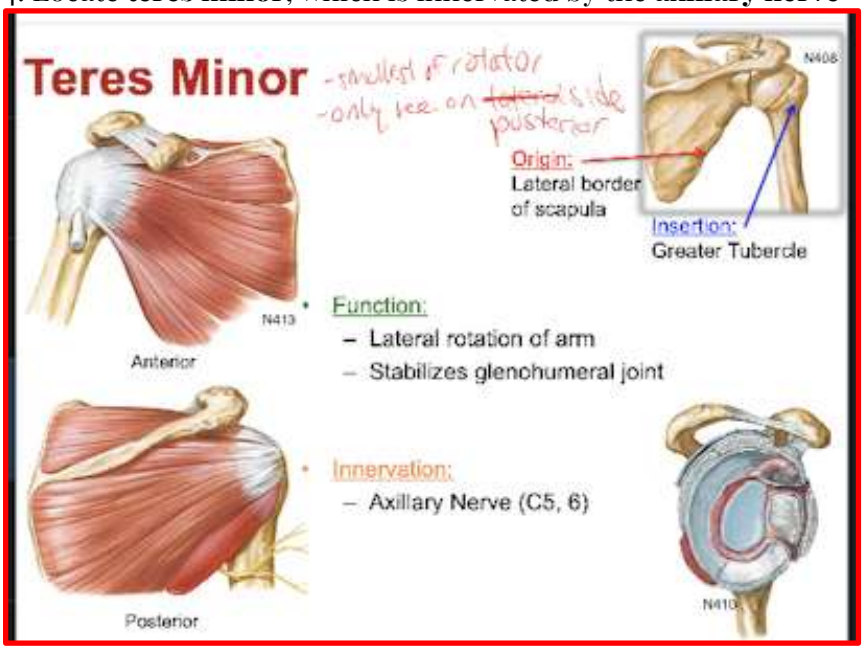
2. ID the **deltoid muscle** (forms the roundness of the shoulder)
   a. Note spine, acromion process (Scapula)
   b. Note deep surface branches of the **axillary nerve** (supplies the deltoid)



3. ID deep muscles to the trapezius:
    a. **Supraspinatus**
    b. **Infraspinatus**
    c. **Teres minor**
    d. Follow their tendons to **head of humerus**
        i. **Rotator cuff muscles (SITS** muscles)

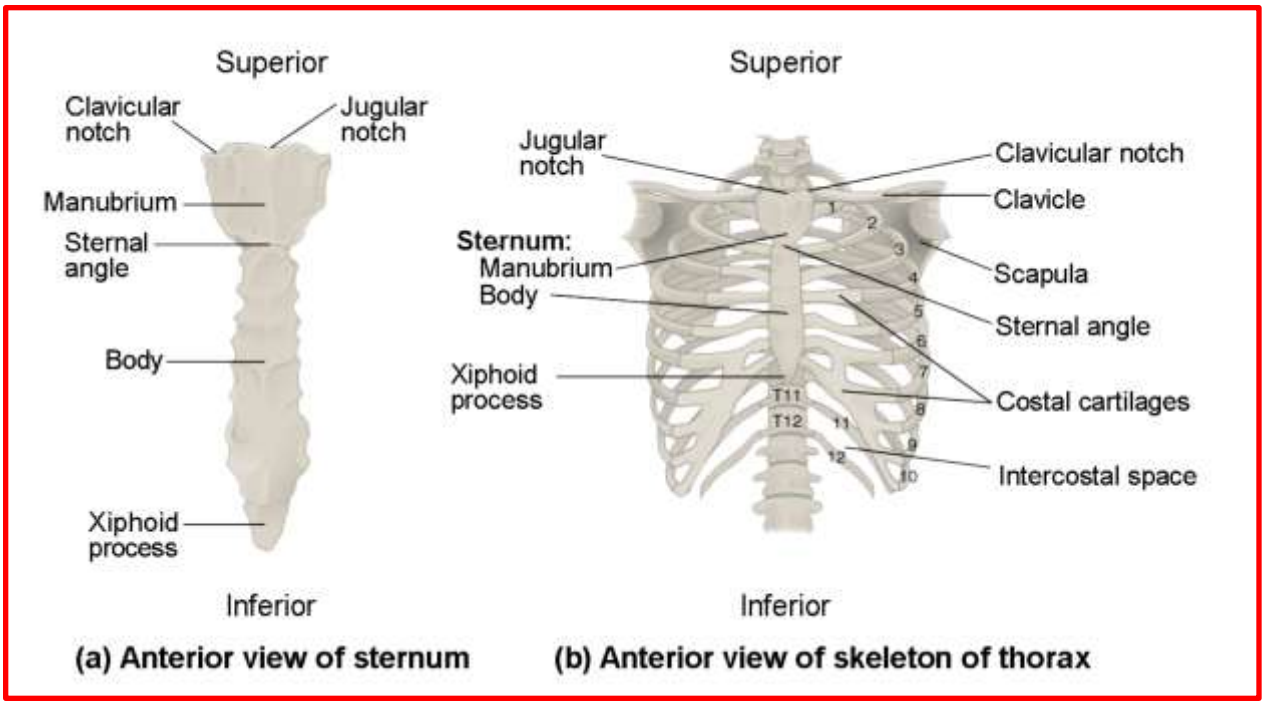


4. Locate **teres minor**, which is innervated by the **axillary nerve**



B. Landmarking Osteology

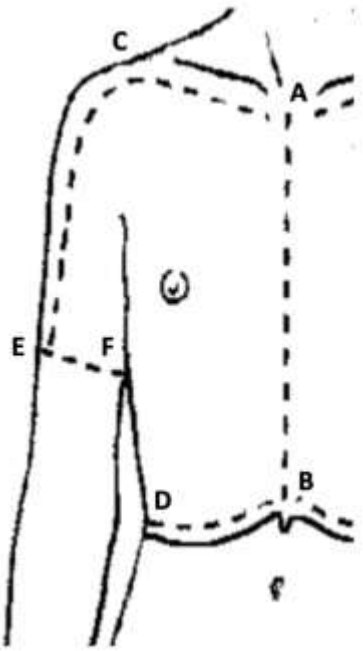
1. Turn cadaver to supine position and locate the following midline structures:
    a. **Sternum**
    b. **Suprasternal Notch** of the **Manubrium**
    c. **Xiphoid Process**
    d. **Sternal angle (of Louis)**, articulation of the **manubrium** with the **sternum body**
        i. **Sternal angle**: level of articulation of the second costal cartilages with the sternum, at the level of the **disc between T4-T5 vertebrae**
2. Locate **clavicle**
    a. Extends from the **sternum** to the **acromion** of the **spine of the scapula**
    b. Observe the following articulations:
        i. Medially with the **manubrium** and the **first costal cartilage** at the **sternoclavicular joint**
        ii. Laterally with the **acromion** at the **acromioclavicular joint**


Superior
Clavicular notch
Jugular notch
Manubrium
Sternal angle
Body
Xiphoid process
Inferior
(a) Anterior view of sternum
Superior
Jugular notch
Clavicular notch
Clavicle
Sternum:
Manubrium
Body
Scapula
Sternal angle
Xiphoid process
T11
T12
Costal cartilages
Intercostal space
Inferior
(b) Anterior view of skeleton of thorax

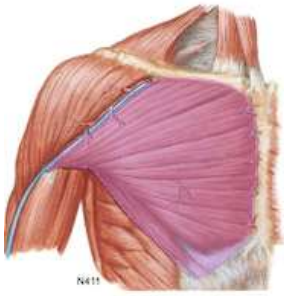
C. Pectoral Region & Axilla

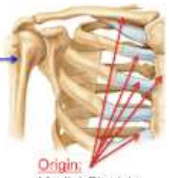
1. Make the following skin incisions as shown in the figure:
   a. Midline from the **suprasternal notch** to the **xiphoid process** (A to B)
   b. From A along the clavicle to the acromion at point C
   c. Then down the arm to point E
   d. Across the arm to point F
   e. Laterally from point B along the costal margin to D in the mid-axillary line

2. Reflect the skin and the superficial fascia laterally to expose the **pectoralis major muscle** and the **deltoid** muscle
   a. Note the **deltopectoral triangle** and the **cephalic vein** coursing within the triangle
   b. The **cephalic vein** disappears in the triangle by piercing the **clavipectoral fascia** and emptying into the **axillary vein**
      i. Preserve the **cephalic vein** as it courses distally down the arm



Pectoralis Major



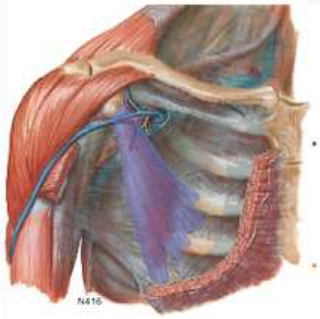


Insertion:
Lateral lip of intertubercular groove

Origin:
Medial Clavicle
Sternum
Upper 6 Ribs and Costal Cartilages

- Function: (think claviprest)
  - Adducts, flexes & medially rotates humerus
- Innervation:
  - Medial & Lateral Pectoral Nerves

Pectoralis Minor

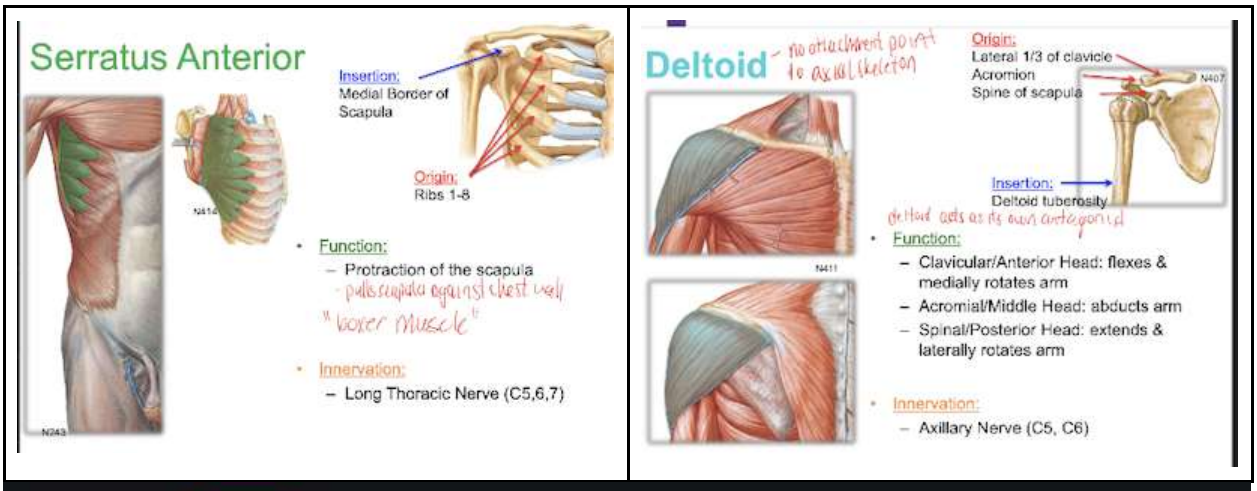


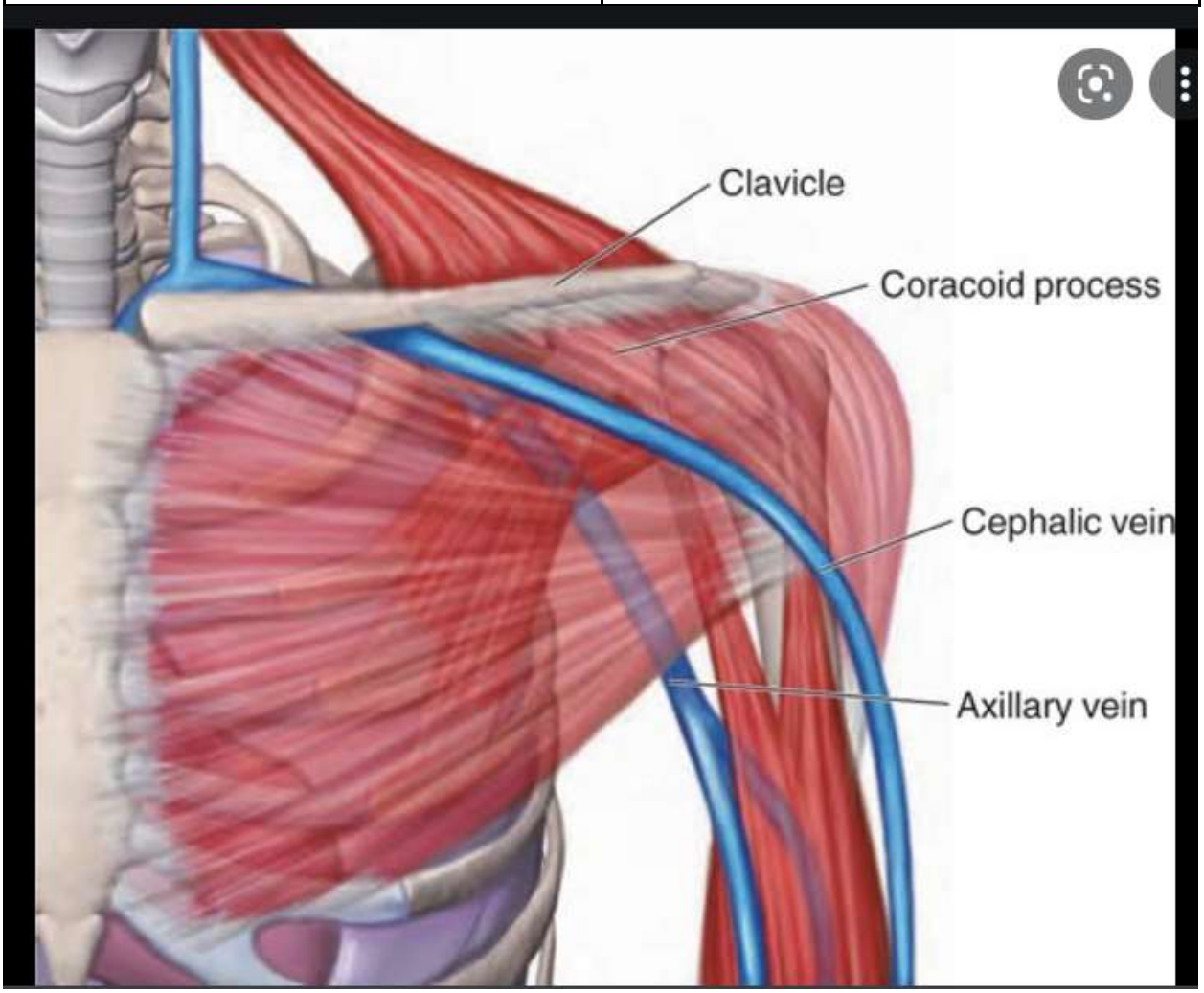


Insertion:
Coracoid Process

Origin:
Ribs 3-5

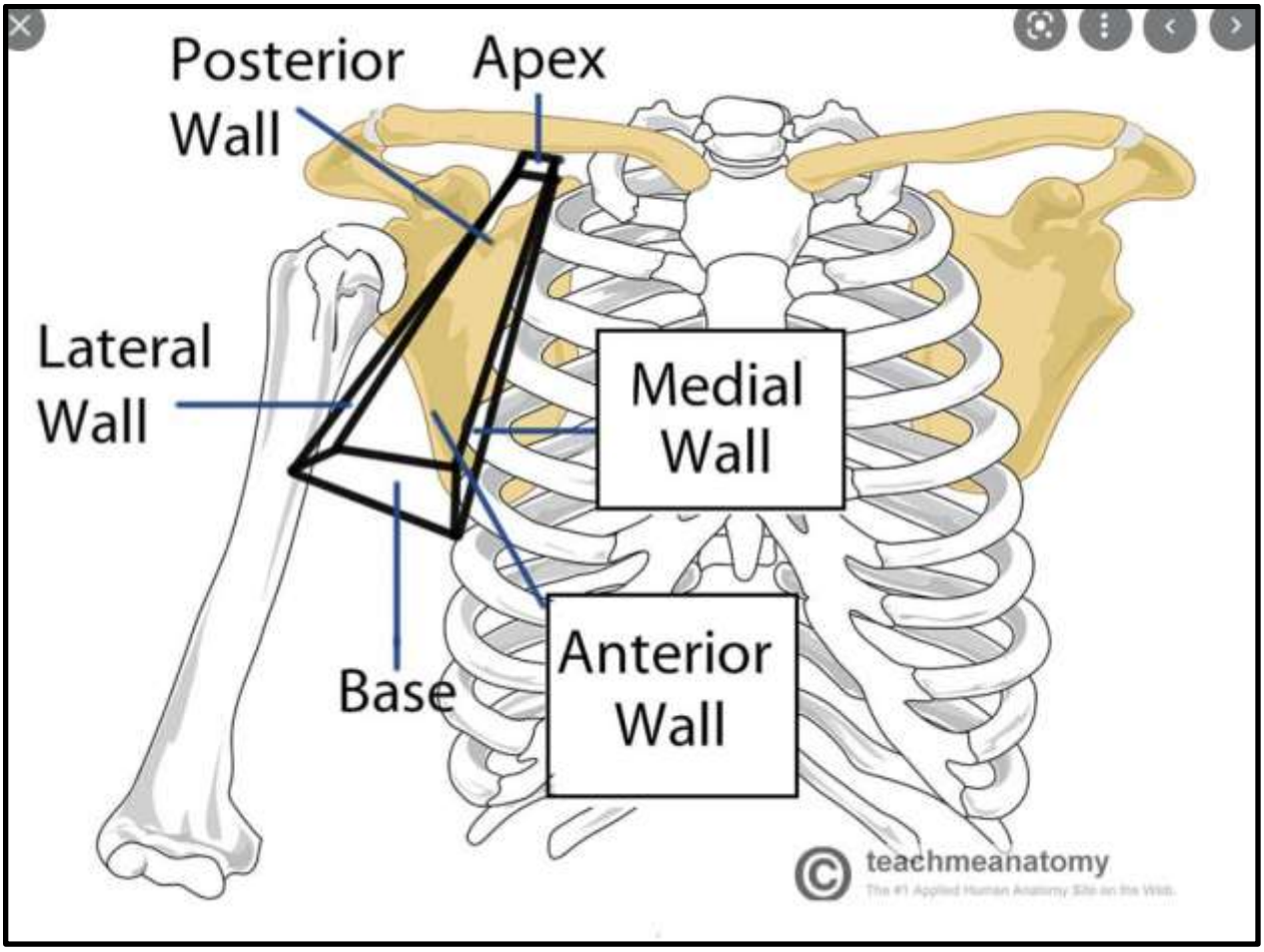
- Function:
  - Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
- Innervation:
  - Medial & Lateral Pectoral Nerves



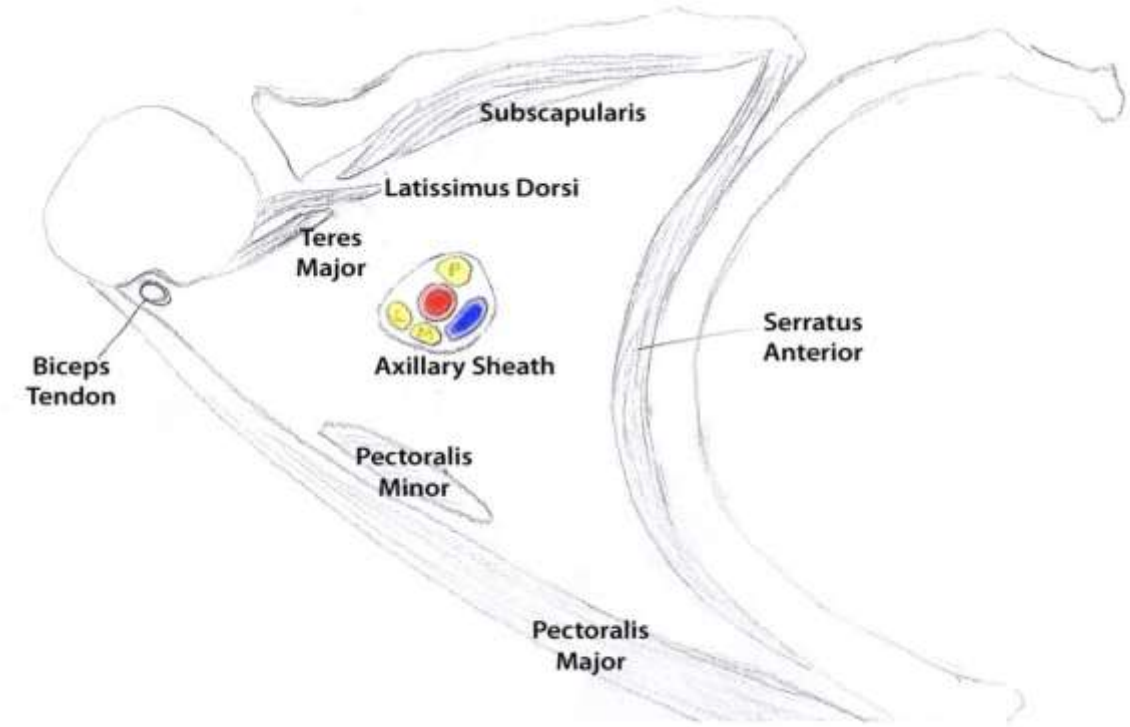


- **Cephalic vein** disappears in the **deltopectoral triangle** by piercing the **clavipectoral fascia** and empties into **axillary vein**

Borders of the Axilla

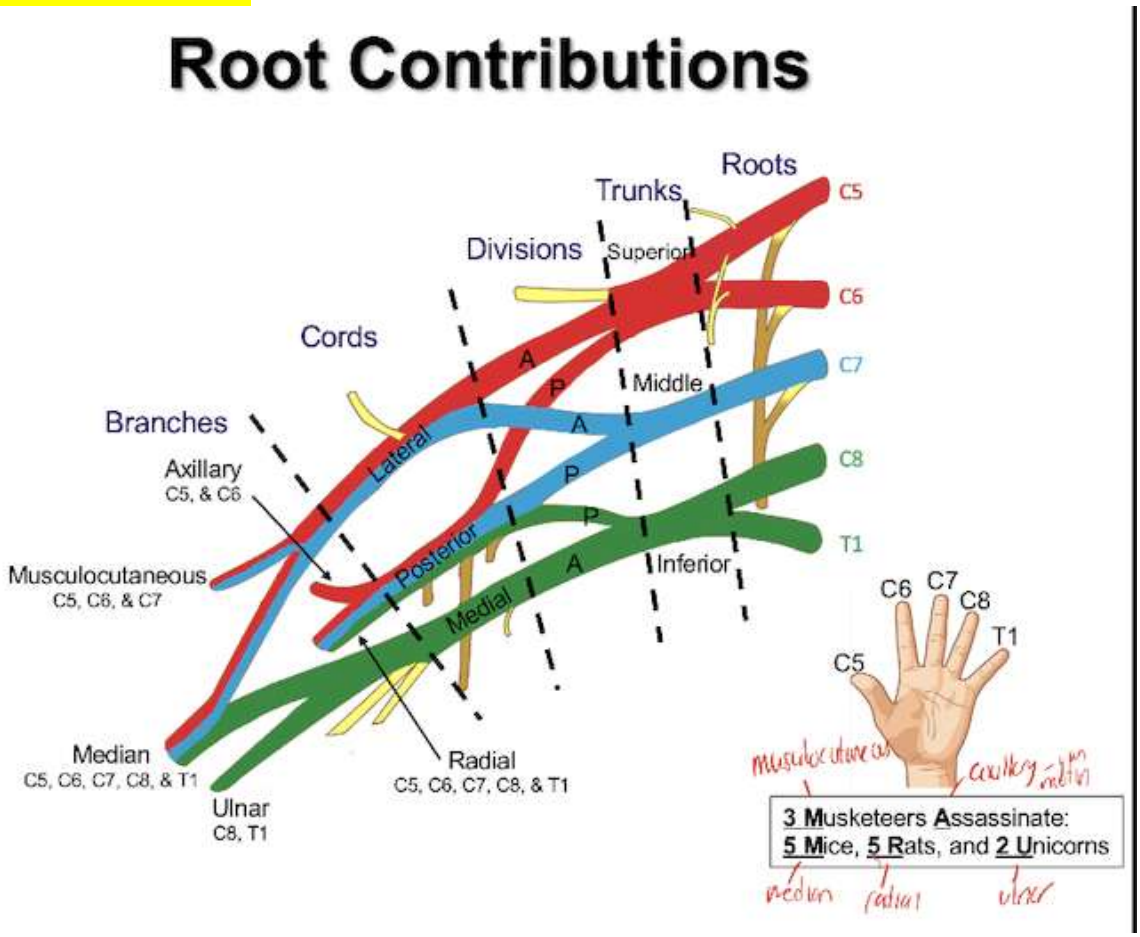


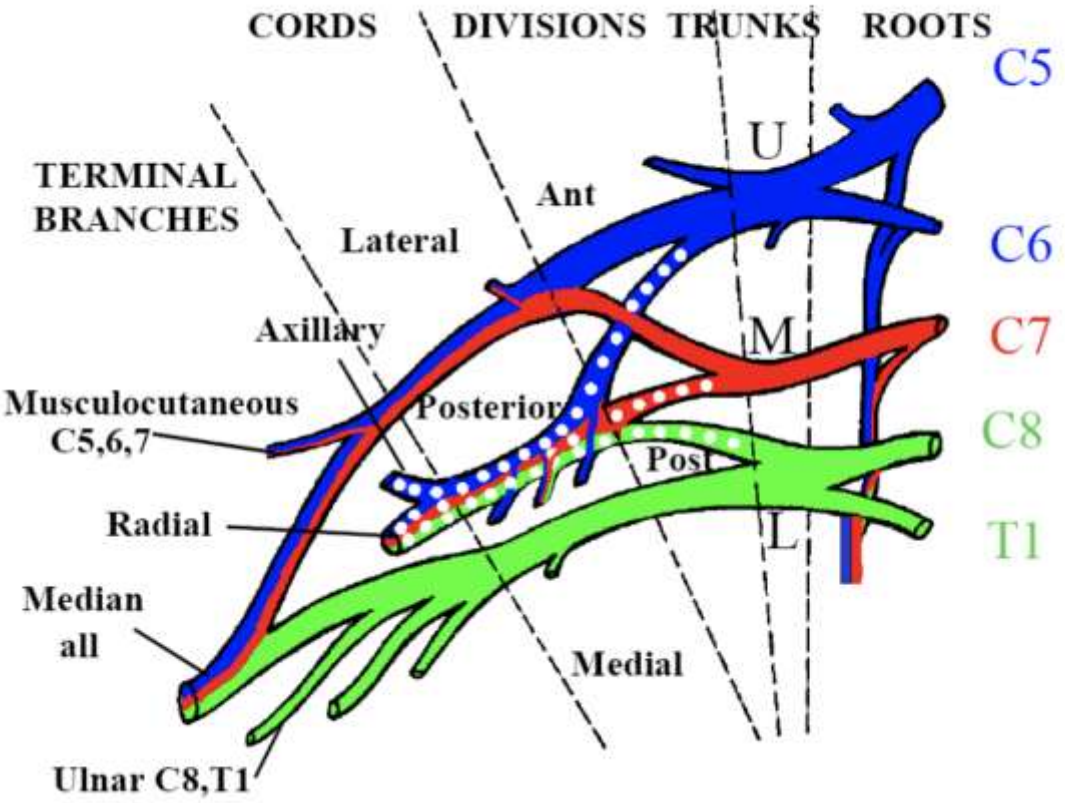
a. Apex:clavicle anteriorly; upper border of scapula posteriorly;rib medially
b. Base:skin and fascia of axilla(armpit)
c. Anterior wall: pectoralis major and minor; clavipectoral fascia
d. Posterior Wall:axillary foldofteresmajorand latissimus dorsi;subscapularis muscle
e. Medial Wall:upper thoracic wall;serratus anterior
f. Lateral wall: intertubercular groove



^Diagram from the Anatomy prep

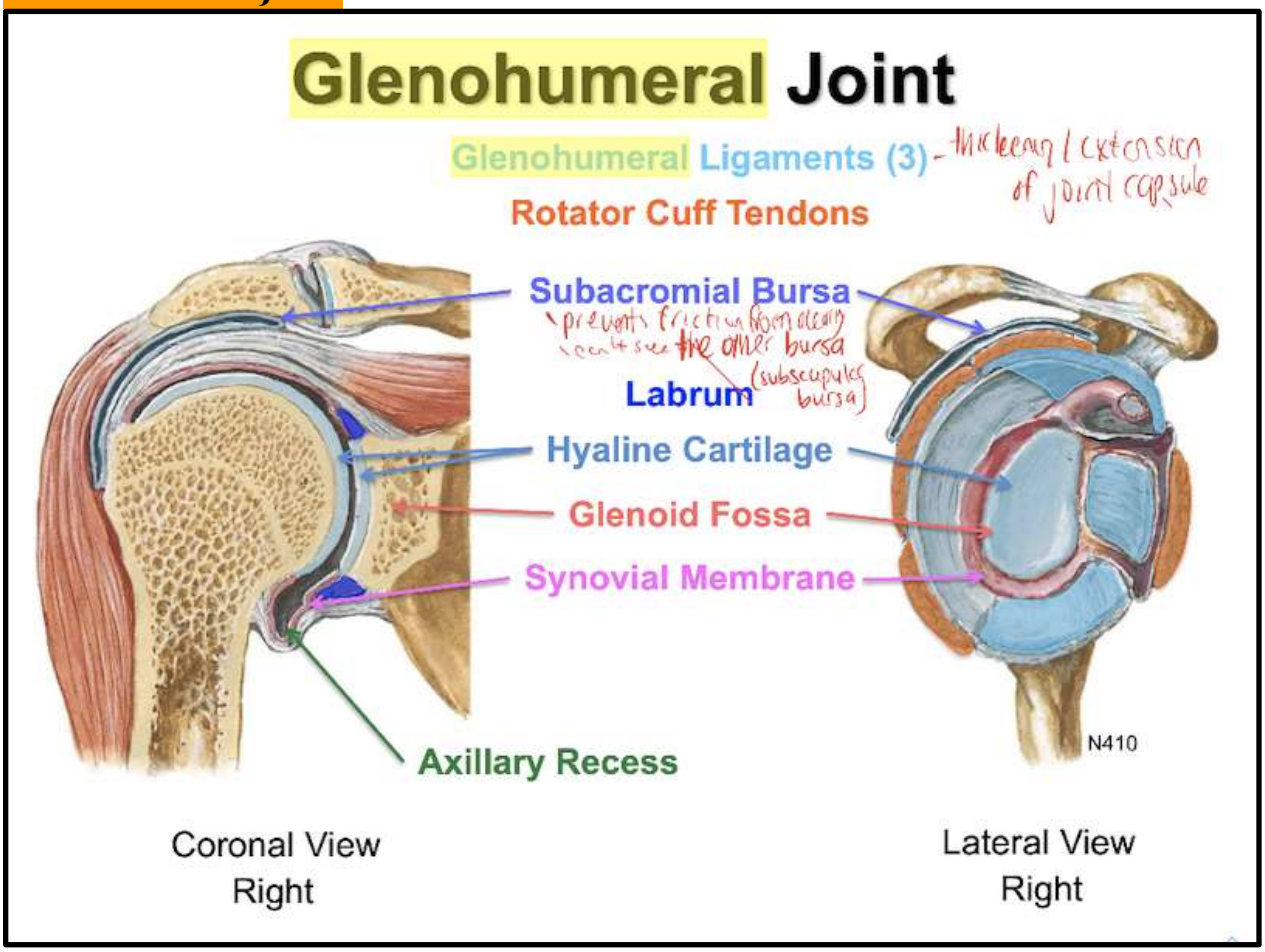
D. Brachial Plexus




CORDS
DIVISIONS
TRUNKS
ROOTS
C5
C6
C7
C8
T1
U
M
L
TERMINAL BRANCHES
Ant
Lateral
Axillary
Posterior
Post
Medial
Musculocutaneous C5,6,7
Radial
Median all
Ulnar C8,T1

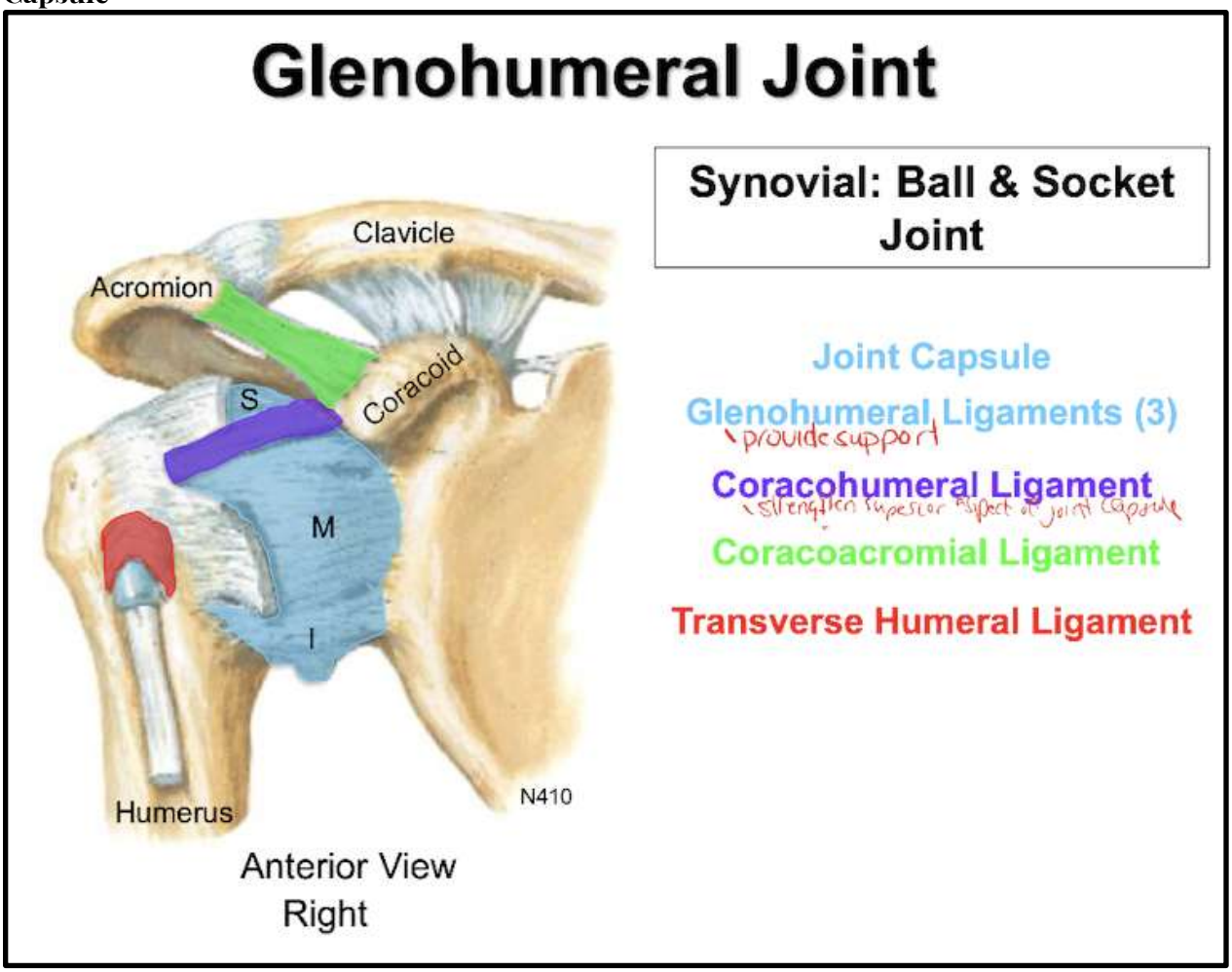
E. Proximal Upper Extremity Joints

Glenohumeral Joint



- Head of Humerus
- Glenoid Fossa (shallow socket)
    - Glenoid Labrum (fibrocartilaginous ring)
        - **Function**: Provide stability and shock absorption within the joint.

Capsule



Major Supporting Ligaments:

- Glenohumeral Ligaments (superior, middle, inferior)
- Coracohumeral Ligament

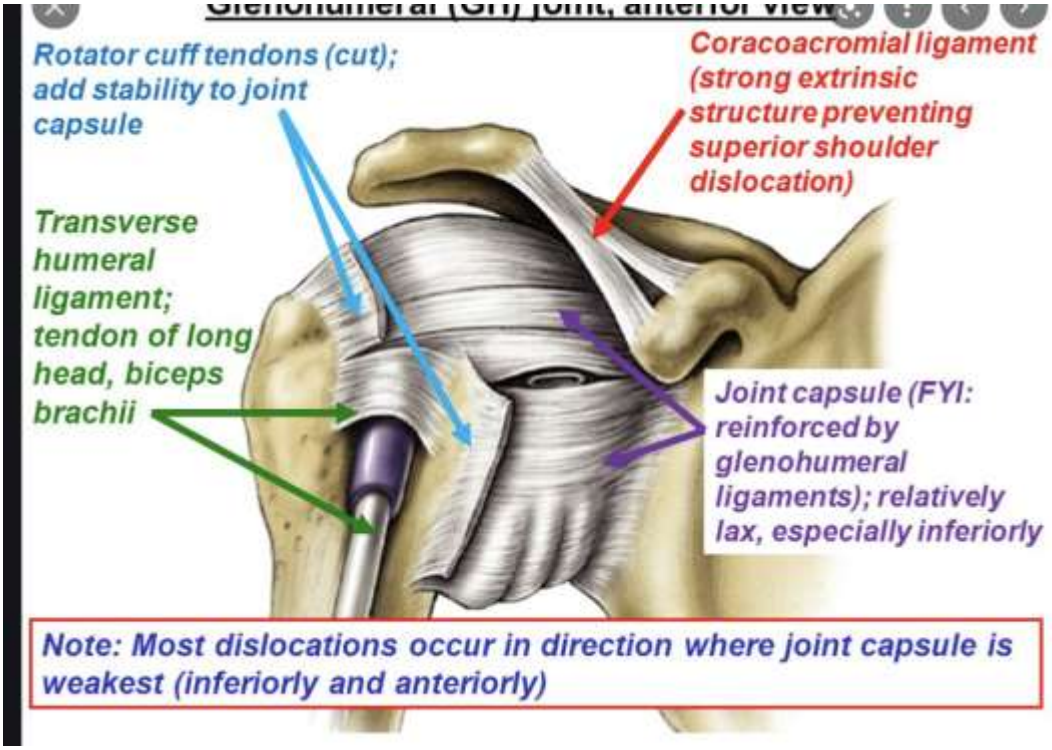
Accessory Ligaments

- Coracoacromial Ligament → forms coracoacromial arch
  - **Function:** osseoligamentous static restraint to superior humeral head displacement.

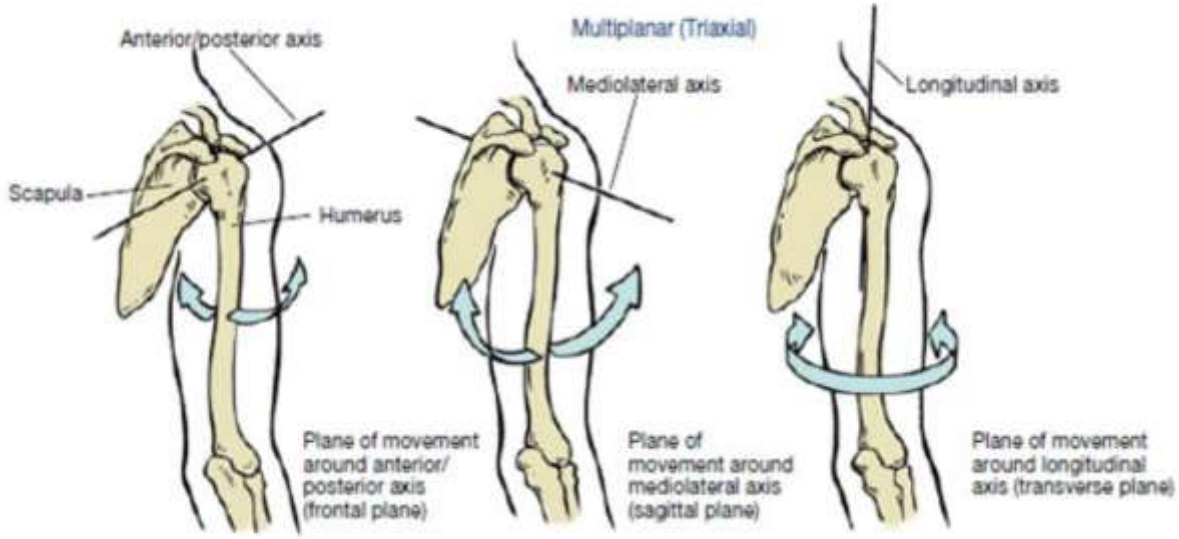
Musculotendinous Reinforcements

- **Weakest region of shoulder joint** and **where do most shoulder dislocations occur**:
  - The weakest area of the shoulder joint capsule is **just anterior to the attachment of the long head of the triceps at the infraglenoid tubercle**, and disloca- tions are most frequent in this inferior anterior region.

-



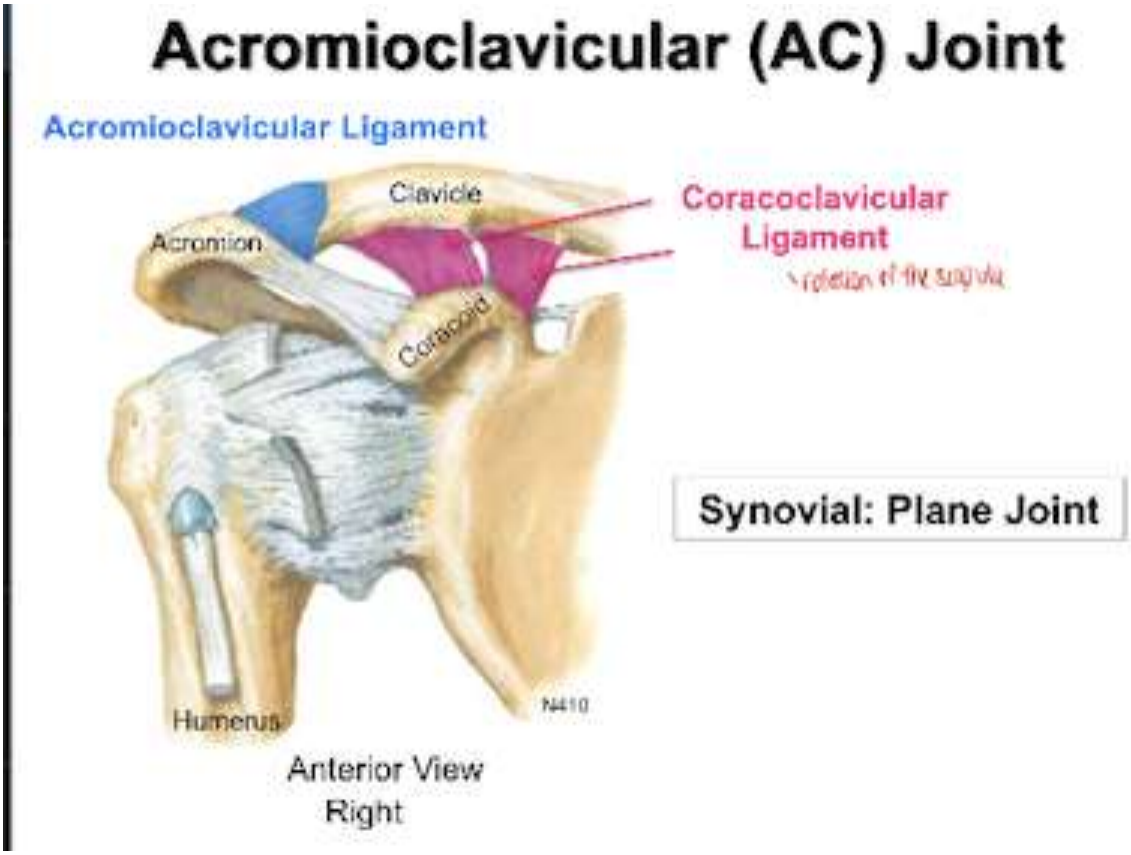
Movements:
Humerus axial movement:



1. **Anterior Posterior Axis**
2. **Mediolateral (Transverse Axis)**
3. **Longitudinal Axis**

Circumduction: movement of a limb or extremity so that the distal end describes a circle while the proximal end remains fixed.

Acromioclavicular (AC) Joint



- **Acromioclavicular ligament**
- **Coracoclavicular ligament**

DAY 2: F- I

F. Vascular Supply to the Axilla
G. Anterior Arm
H. Neurovascular Supply to the Shoulder
I. Posterior Arm